

CHAPTER 7

LOOPING STATEMENTS

Looping Statements / Looping Techniques are special statements used to execute a set of instructions repeatedly for a desired number of times.

Loops are basically used for repeated execution.

TYPES OF LOOPING STATEMENTS IN JAVA:

1. **while loop**
2. **do while loop**
3. **for loop**
4. **nested loop**
5. **for-each loop**

Parts of a Loop

1. **Initialization:** Assign initial value to a variable (int a = 1)
2. **Condition:** It is checked to continue with the loop or end the loop (a<=10)
3. **Updation:** Update the loop variable in every iteration (a++)

Loop Execution Concept

- Looping is often called backtracking technique.
- Condition is checked to either continue the loop or terminate and exit the loop.

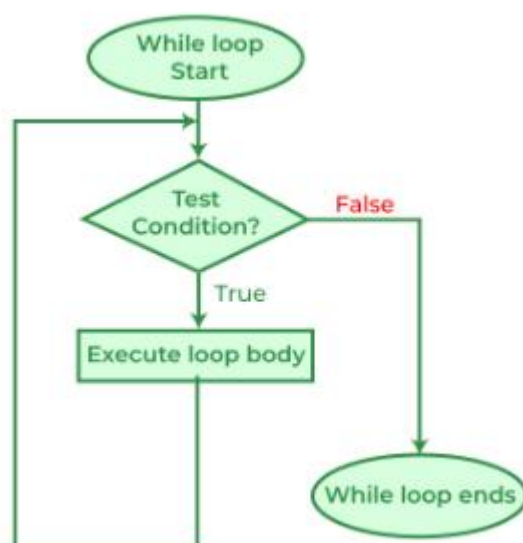
While Loop

- Entry-controlled loop, 'while' is a keyword.
- Loop executes only if the condition is true.

Syntax:

```
1 Initialization;  
2 while (condition) {  
3     // statements  
4     updation;  
5 }  
6
```

FLOW DIAGRAM:



Example:

```
1  class ExampleWhileLoop
2  {
3      public static void main(String [] args)
4      {
5          int a = 1; //INITIALIZATION
6          while(a<=10) //CONDITION
7          {
8              System.out.println(a);
9              a++; //UPDATION
10         }
11     }
12 }
```

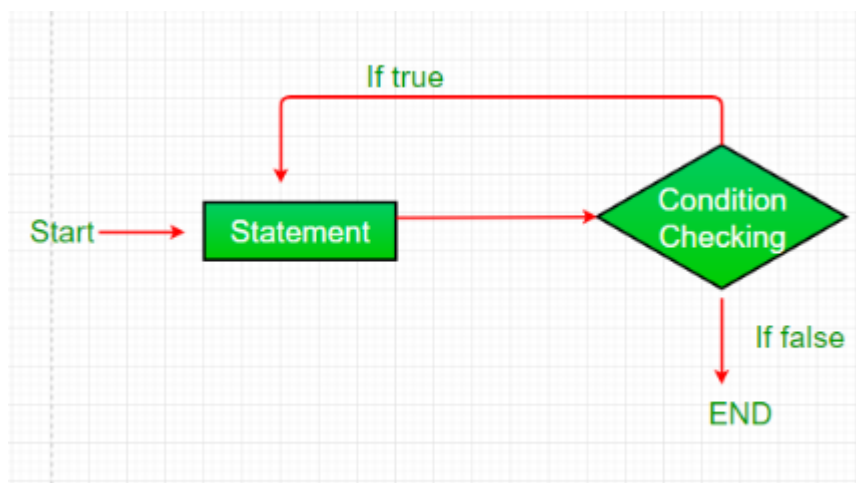
Do While Loop

- Exit-controlled loop, 'do' and 'while' are keywords.
- Executes at least once, even if the condition is false

Syntax:

```
1  Initialization;
2  do {
3      // statements
4      updation;
5  } while (condition);
6
```

FLOW DIAGRAM:



Example:

```
1  class ExampleDoWhileLoop
2  {
3      public static void main(String [] args)
4      {
5          int a = 1; //INITIALIZATION
6          do
7          {
8              System.out.println(a);
9              a++; //UPDATION
10         }while(a>=5); //CONDITION
11     }
12 }
```

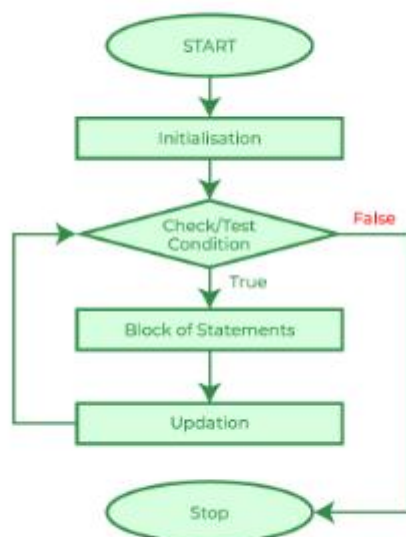
For Loop

- Most commonly used loop.
- Entry-controlled, 'for' is a keyword.
- All three parts (initialization, condition, updation) are in a single line.

Syntax:

```
1 for (initialization; condition; updation) {  
2     // statements  
3 }  
4
```

FLOW DIAGRAM:



Example:

```
1 class ExampleForLoop
2 {
3     public static void main(String [] args)
4     {
5         for(int i=1; i<=10; i++) //INITIALIZATION, CONDITION AND UPDATION
6         {
7             System.out.println(i);
8         }
9     }
10 }
```

Advantages of for loop:

- Compact syntax
- Code optimization
- Better performance
- Programmer-friendly

Nested Loops

- Writing one loop inside another loop's body.
- For each iteration of the outer loop, the inner loop executes completely.

Example:

```
1 class ExampleNestedForLoop
2 {
3     public static void main(String [] args)
4     {
5         for(int i=1; i<=10; i++) //INITIALIZATION, CONDITION AND UPDATION
6         {
7             for(int j=1; j<=5; j++) //INITIALIZATION, CONDITION AND UPDATION
8             {
9                 System.out.println(" * ");
10            }
11        }
12    }
13 }
14
```

Infinite Loop

- A loop that never terminates unless forcibly broken using break.

Using While Loop

```
1  class ExampleInfiniteLoop
2  {
3      public static void main(String [] args)
4      {
5          while(true) //INFINITE LOOP, ALWAYS TRUE
6          {
7              System.out.println(" * ");
8          }
9      }
10 }
```

Using For Loop

```
1  class ExampleInfiniteLoop
2  {
3      public static void main(String [] args)
4      {
5          for( ; ; ) //INFINITE LOOP, ALWAYS TRUE
6          {
7              System.out.println(" * ");
8          }
9      }
10 }
```

Difference between for loop and while loop:

for Loop	while Loop
Initialization may be either in the loop statement or outside the loop.	Initialization is always outside the loop.
Once the statement(s) is executed then increment is done.	The increment can be done before or after the execution of the statement(s).
It is normally used when the number of iterations is known.	It is normally used when the number of iterations is unknown.
Condition is a relational expression.	The condition may be an expression or non-zero value.
It is used when initialization and updation of conditions are simple.	It is used for complex initialization.
For loop is entry controlled loop.	While loop is also entry controlled loop.
Syntax: for (init ; condition ; iteration) { statement(s); }	Syntax: while (condition) { statement(s); }
The for loop is used when the number of iterations is known.	The while loop is used when the number of iterations is unknown.